UNIT 1: RECAP

Introduction to Language & Natural Language Processing

Language & Its Evolution

Before we teach machines to understand language, let's understand how human language itself came to be.

What Is Human Language?

A SIMPLE ANALOGY

Think of language as a shared code.

When you say the word **“apple”**, a picture of the fruit pops into the listener's head — because both of you agreed, long ago, that this sound/symbol means that object. Language is simply a system of sounds, signs, and symbols that lets one mind transfer an idea into another mind.

Why it matters for NLP:

If a computer can learn this same shared code — which sounds/words map to which meanings — it can read, understand, and even generate human language.

KEY DEFINITION

Language

A structured system of communication — spoken, written, or signed — made of vocabulary (words) and grammar (rules for combining them) that a community agrees to share.

Two forms of language:

- Spoken (speech / audio)
- Written (text)

NLP works with both.

Evolution of Natural Language

Human language has evolved through five distinct stages — each expanding how thought is shared, stored, and eventually understood by machines.



NLP Is the Newest Stage of This Evolution



ONE MESSAGE, FIVE STAGES — "COME TO THE CLASSROOM"

- You say it — your friend hears it.
- You write it — your friend reads it.
- You print it — distributed to 100 students.
- You send it on WhatsApp — received instantly.
- You tell an AI—it understands the intent and generates: "Hi, please come to the classroom."

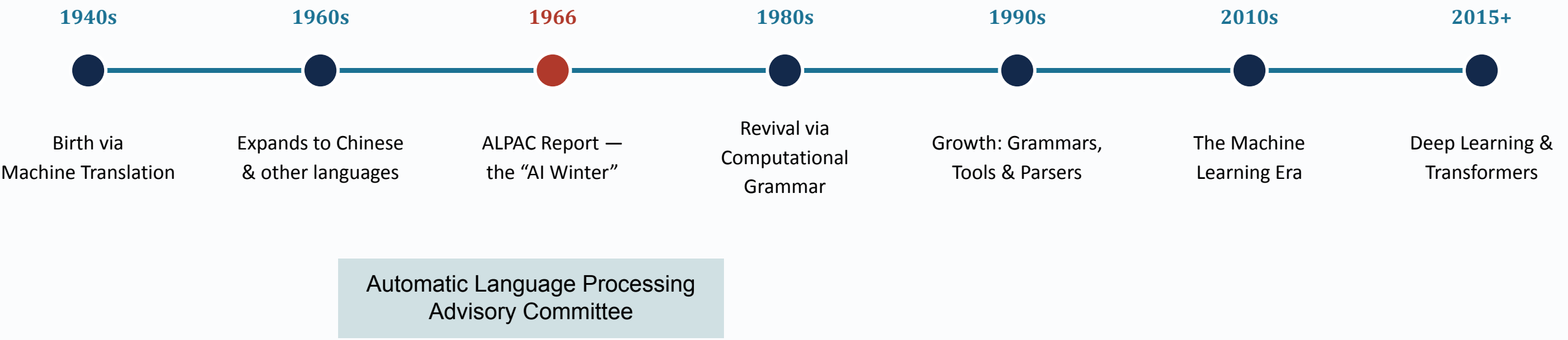
Humans spent thousands of years learning to speak, write, print, and digitize. Now machines are learning to read, understand, interpret, and generate language too.

History of NLP

A field that nearly died in 1966 — then was reborn twice: once through linguistics, and again through machine learning and deep learning.

The Journey of NLP—Full Timeline

From a 1940s translation problem to today's transformer-powered systems — NLP's path was anything but a straight line.



The Machine Learning Era

The Five Classical ML Algorithms

1

Naïve Bayes

Probabilistic classifier; a long-time favorite for spam & text classification.

2

Logistic Regression

Simple, interpretable linear model for binary/multi-class text labels.

3

Support Vector Machine

Finds the optimal boundary between classes in high-dimensional text feature space.

4

Latent Dirichlet Allocation

Unsupervised topic modeling — discovers hidden themes across documents.

5

Hidden Markov Model

Models sequences with hidden states — used for POS tagging & speech.

Deep Learning & the Age of Transformers

RNN

Recurrent Neural Network — processes text sequentially, one word at a time.

LSTM

Long Short-Term Memory — an RNN variant that remembers longer-range context.

GRU

Gated Recurrent Unit — a lighter, faster alternative to LSTM.

Transformers

Attention-based architecture (2017) — powers virtually all modern LLMs.

Autoencoders

Learn compressed representations of text for reconstruction & feature learning.

2017 was the pivot point: Transformers made possible everything from BERT to modern large language models like GPT and Claude.

Why Human Language Is Hard for Machines ?

Ambiguity

One sentence can be read in more than one way.

"I saw the man with a telescope." — who has the telescope?

Meaning 1:
I used a telescope to see the man.

Meaning 2:
I saw a man who was carrying a telescope.

A human can use the surrounding situation to understand the intended meaning.

But a computer has to ask:

"Who has the telescope?"

This is difficult because the words themselves don't provide enough information.

Context-Dependence

The same words can mean different things in different situations.

"It's cold in here" can be a fact — or a request to close the window.

"It's cold in here."

It could simply mean: "The temperature is low."

But imagine someone sitting next to an open window.

"It's cold in here."

They might actually be requesting:

"Please close the window."

The person doesn't directly say **"Close the window."**

A human understands the hidden intention from the situation.

This is called **context-dependent meaning**.

Why Human Language Is Hard for Machines

Creativity

People constantly invent new words, slang & idioms.

“ghosting”, “break the ice” — new meanings that aren't in any old dictionary.

“He ghosted me.”

usually means: He suddenly stopped communicating with me.

“She told a joke to break the ice.”

The person isn't actually breaking physical ice. 🧊

It means:To make people feel more comfortable in a social situation.

So NLP systems need to understand idioms and evolving language, not just individual words.

Structure & Grammar

Word order and grammar rules vary widely across languages.

English: Subject-Verb-Object. Other languages order words very differently.

English commonly follows:

Subject + Verb + Object

English:

Ravi eats mangoes.

Ravi → Subject
eats → Verb
mangoes → Object

But different languages can use different word orders.

For example, Hindi can use:

Ravi aam khata hai.

This becomes even more challenging for NLP because there are **thousands of human languages**, each with its own:

- Grammar
- Word order
- Writing system
- Vocabulary
- Expressions
- Cultural context

Natural Language Processing (NLP) is the field built to solve exactly these problems — giving computers the ability to read, understand, interpret, and generate human language.

Human Language → Problems → NLP Solutions

Ambiguity

NLP learns the possible meanings a word or phrase can take.

Context

NLP uses the surrounding words and situation to pick the right meaning.

Creativity

NLP keeps learning new words, slang, and expressions as language evolves.

Grammar

NLP learns the structure and rules that hold a language together.

Teaching a Computer Isn't Just Teaching a Dictionary

Classroom analogy: teaching a new student the word “apple.”



Fruit

“I ate an apple.”

The literal, everyday meaning.



Company

“Apple released a new iPhone.”

Same word, a brand name instead.



Idiom

“That’s the apple of my eye.”

A figure of speech — meaning “cherished.”

So teaching a computer language is much more than teaching it a dictionary — that’s the fundamental challenge NLP tries to solve.

Foundations of NLP

How ML, DL, and NLP Relate

Natural Language Processing (NLP)

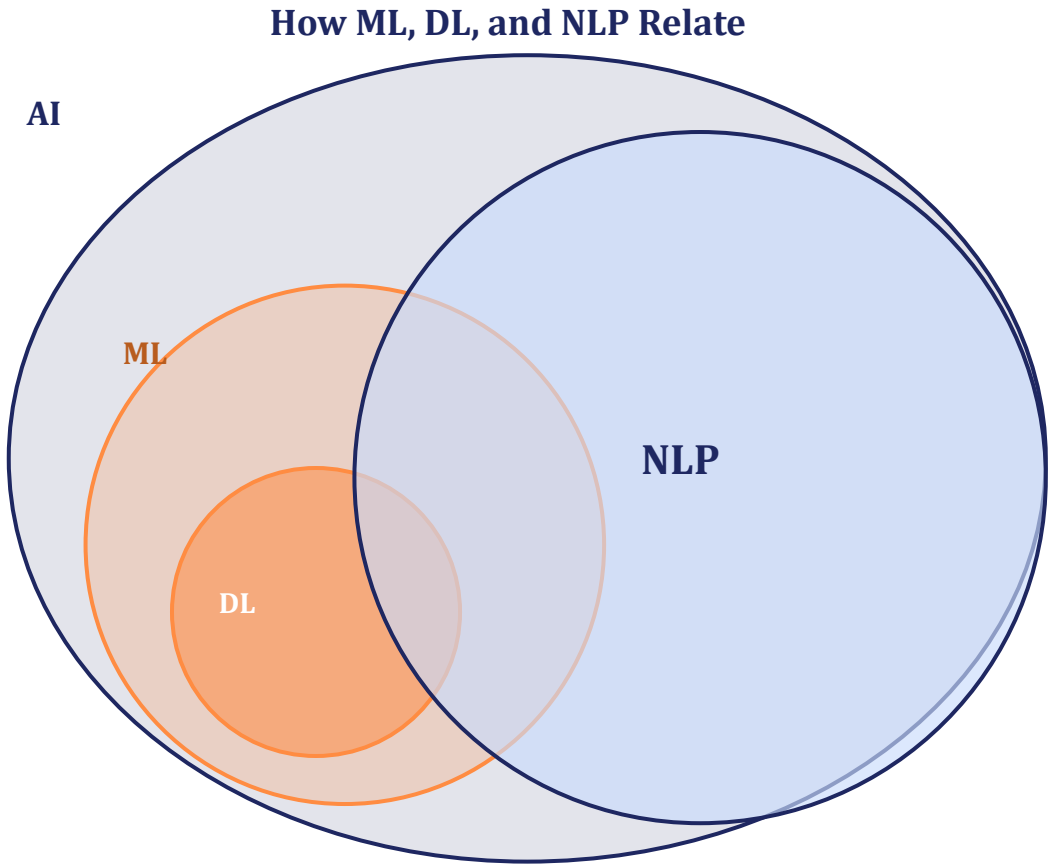
Natural Language Processing (NLP) is a branch of data science that consists of systematic processes for analyzing, understanding, and deriving information from the text data in a smart and efficient manner

Computational Linguistics

Roughly, the intersection of computer science and linguistics — methods for analysis/synthesis tasks, and models that explain linguistic phenomena using knowledge or statistics.

Linguistics

The study of natural language(s) in terms of form, meaning, and context.



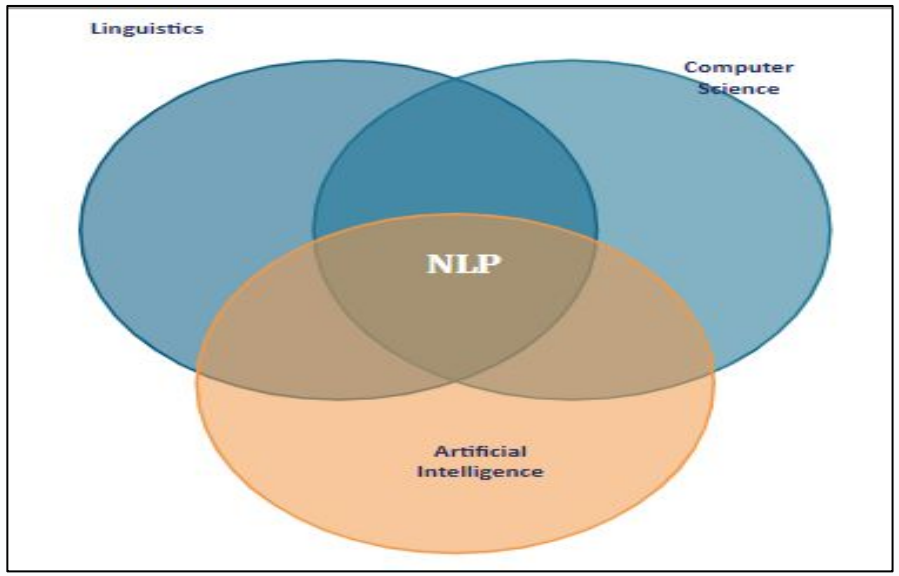
What Is Natural Language Processing?

IBM'S DEFINITION

"Natural language processing strives to build machines that understand and respond to text or voice data — and respond with text or speech of their own — in much the same way humans do."

WORKING DEFINITION

"The ability of a computer program to understand human language as it is spoken and written — referred to as natural language. It is a component of artificial intelligence."



IN PLAIN TERMS

NLP = AI + Computer Science + Human Language

- Handles text (written) and speech (spoken)
- Combines rule-based grammar with statistical & deep-learning models
- Goal: bridge human communication and machine computation

EXAMPLE 1 — MACHINE TRANSLATION

How a Simple Translation System Works

INPUT (ENGLISH)

“Good morning”

→

OUTPUT (GUJARATI)

“સુપ્રભાત”

L

Linguistics

Helps understand what “Good morning” actually means.

CL

Computational Linguistics

Represents that meaning in a form a computer can work with.

NLP

NLP

Defines the task itself: English → Gujarati translation.

ML

Machine Learning

Learns translation patterns from large amounts of bilingual text.

DL

Deep Learning

Uses neural networks / transformers to learn complex relationships between languages.

AI

AI

Provides the overall intelligent system that ties every layer together.

EXAMPLE 2 — SENTIMENT ANALYSIS

How a Gujarati Sentiment Classifier Works

STUDENT WRITES (GUJARATI)

“આ ફિલ્મ ખૂબ જ સરસ છે.”

→

TRANSLATION

“This movie is very good.”

L

Linguistics

We need to understand Gujarati grammar, words, meaning, and context.

CL

Computational Linguistics

We develop computational methods for representing and processing Gujarati.

NLP

NLP

The system processes the Gujarati sentence and determines its sentiment.

ML

Machine Learning

The model learns from many examples of positive and negative Gujarati sentences.

DL

Deep Learning

A neural network learns complex language patterns in the text.

AI

AI

The overall system ties every layer together to perform the intelligent task.

Final Result: Sentiment = Positive 😊

So What Does NLP Do?

NLP = Natural Language Processing

A field of AI + Linguistics + Computer Science that teaches computers to work with human language.

HUMAN ABILITY →	NLP TASK
Read text	**Text processing**
Understand words	**Tokenization / semantics**
Understand grammar	**Syntactic analysis**
Understand meaning	**Semantic analysis**
Understand context	**Contextual modelling**
Understand emotion	**Sentiment analysis**
Translate languages	**Machine translation**
Answer questions	**Question answering**
Talk with humans	**Chatbots**
Generate text	**Natural Language Generation**
Understand speech	**Speech recognition**

Types of NLP: Understanding vs. Generation

NLU & NLG

The two halves of how machines hold a conversation

NLU

Natural Language Understanding

Human language → structured meaning

Understand

⇄

NLG

Natural Language Generation

Structured meaning → human language

Speak / Generate

NLU: Understanding Meaning and Intention

The part of NLP that maps human input into a useful, structured representation.

USER SAYS

“Book me a flight to Mumbai tomorrow.”

INTENT	DESTINATION	DATE	USER ACTION
Book a flight	Mumbai	Tomorrow	Make a booking

MAPPED INTO A STRUCTURED REPRESENTATION

```
Intent = Flight_Booking
Destination = Mumbai
Date = Tomorrow
```

The computer converts human language → structured information.

What Does NLU Actually Analyze?

Five aspects of language NLU can pull apart from a single sentence.

Words

“Book a flight”

Key words: book, flight

Grammar

“Ravi eats mangoes.”

Ravi = Subject · eats = Verb · mangoes = Object

Meaning

“The movie was fantastic.”

Expresses a positive opinion

Intent

“Can you show me today’s weather?”

Intent: weather information request

Entities

“Book a ticket to Delhi.”

Delhi = Location · ticket = Object

NLG: Turning Data Into Natural Language

Takes structured information and turns it into human-readable sentences.

STRUCTURED DATA

```
Temperature = 30°C
Condition = Sunny
Location = Mumbai
```

→

NATURAL LANGUAGE OUTPUT

“The temperature in Mumbai is 30°C and it is sunny today.”

In short: NLG converts computer information → human-readable language.

What Does NLG Involve?

Two steps turn raw data into a grammatically correct sentence.

1 Text & Sentence Planning

The computer decides: what should I say, and how should it be organized?

```
Temperature = 30°C
Weather = Sunny
Humidity = 70%
```

→ **“Today is sunny in Mumbai, with a temperature of 30°C.”**

2 Text Realization

The computer decides: how should I actually write the sentence?

```
Mumbai + temperature + 30°C + sunny
```

→ **“Mumbai is experiencing sunny weather with a temperature of 30°C.”**

NLU + NLG TOGETHER

A Chatbot Answering a Question

“What is the weather in Mumbai today?” — walked through both halves of NLP.

STEP 1 · NLU

Understands the question:
Intent = Weather
Location = Mumbai
Date = Today

→

STEP 2 · DATA LOOKUP

Weather service provides:
Temperature = 30°C
Condition = Sunny

→

STEP 3 · NLG

Generates the reply: “The weather in Mumbai is sunny today, with a temperature of 30°C.”

NLU vs NLG

NLU	NLG
Understands human language	Produces human language
Human ⟶ Computer	Computer ⟶ Human
Extracts meaning	Expresses meaning
Finds intent	Creates response
Finds entities	Creates sentences
Text ⟶ Structured representation	Structured representation ⟶ Text
“Book a flight to Delhi” ⟶ booking request	Flight booked + Delhi ⟶ “Your flight to Delhi has been booked.”

Need & Applications of NLP

✳ The Need for NLP ✳

Why NLP is Needed	Meaning	Example
1. Explosion of Unstructured Text 	Huge amount of text is created every day but it is not organized.	A company gets 10,000 customer reviews. NLP can find common complaints.
2. Automation at Scale 	NLP can process thousands or millions of documents automatically.	Instead of a person reading 1 lakh emails, NLP can identify Spam / Important / Promotional emails.
3. Natural Human-Computer Interaction 	Allows humans to communicate with computers using normal language.	You say "What is the weather today?" to Siri / Google Assistant, it gives the answer.
4. Better Decision-Making 	NLP extracts useful information from large amounts of text to help make decisions.	A company analyzes reviews and finds "battery problem" is the most common complaint.
5. Breaking Language Barriers 	NLP can translate one language into another.	"How are you?" → आप कैसे हैं?
6. Accessibility 	Helps people interact with technology through speech and text.	Speech-to-text: A person speaks → computer converts it into written text.

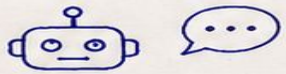
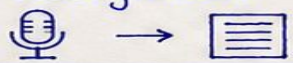


NLP helps computers:

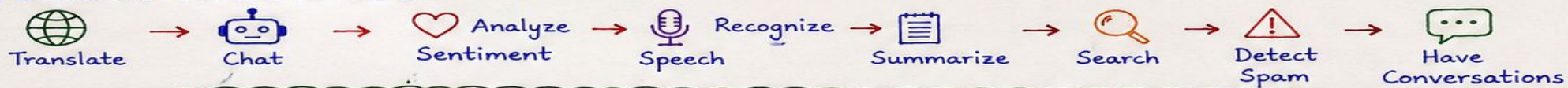
Read → Understand → Analyze → Translate → Respond

Human Language → NLP → Useful Information / Action

✳ Applications of NLP ✳

Application	What it does	Example
1. Machine Translation 	Converts text from one language to another.	English: "Good morning" ↓ Hindi: "सुप्रभात"
2. Chatbots & Virtual Assistants 	Understands user questions and gives responses.	User: "What are your working hours?" Bot: "We are open 9 AM – 6 PM."
3. Sentiment / Opinion Mining 	Identifies whether an opinion is positive, negative or neutral.	"This phone is excellent." → Positive
4. Speech Recognition 	Converts spoken language into text.	You say: "Call Mom" → Phone converts speech into text / command.
5. Text Summarization 	Converts a long text into a short summary.	10-page news article → 3-4 important sentences.
6. Information Retrieval 	Finds relevant information from a large collection of documents.	Search "best laptop for students" → Search engine finds relevant pages.
7. Spam & Fraud Detection 	Identifies unwanted, suspicious or fraudulent messages.	"Congratulations! You won ₹10 lakh. Click here." → Spam
8. Discourse & Dialog Systems	Maintains context across multiple conversations.	User: "Book a flight to Delhi." Bot: "For which date?" User: "Tomorrow."

Easy Way to Remember:



NLP makes computers understand, process and generate human language!

Discourse & Dialog Analysis

DISCOURSE ANALYSIS

Studies how sentences connect to form coherent, meaningful text — resolving **what pronouns refer** to and how ideas link across sentences.

"Rahul bought a laptop. He loves it."

"**He**" refers back to "Rahul" — this is called coreference resolution.

Used in: document summarization, essay coherence scoring, machine translation of long text.

DIALOG ANALYSIS

Analyzes multi-turn conversations to track intent, context & state across exchanges — the core of chatbots.

User "Book me a flight to Delhi tomorrow."

Intent: Book_Flight

Bot "Sure — what time would you like to depart?"

Slot needed: time

User "Morning, please."

Slot filled: time = morning

Bot "Booking your morning flight to Delhi."

Action: Confirm_Booking

Opinion Mining (Sentiment Analysis)

Opinion mining automatically identifies and extracts subjective information — attitudes, emotions, and opinions — expressed in text, and classifies it as positive, negative, or neutral.

NEGATIVE

"The battery life is terrible and it overheats constantly."

NEUTRAL

"The phone arrived on Tuesday as scheduled."

POSITIVE

"Absolutely love the camera quality — best purchase this year!"

Used for:

Brand monitoring · Product review analysis · Social-media trend tracking · Stock-market sentiment · Customer feedback triage

Four Core Functions of NLP

1

Text Classification

Assigning labels to texts — e.g. recognizing emotion or discovering intent.

2

Text Extraction

Analyzing and extracting keywords that capture the essence of a text.

3

Machine Translation

Automated translation of text from one language to another.

4

Natural Language Generation

Collecting unstructured data, analyzing it, and generating quality content.

Example: Customer Review

Suppose a customer writes:

“The phone is excellent and the camera is amazing.”

- Text Classification: → Positive Review
- Text Extraction: → phone, excellent, camera, amazing
- Machine Translation: → Translate the review into Hindi
- NLG: → Generate a response: “Thank you for your positive feedback!”

Linguistic Foundations

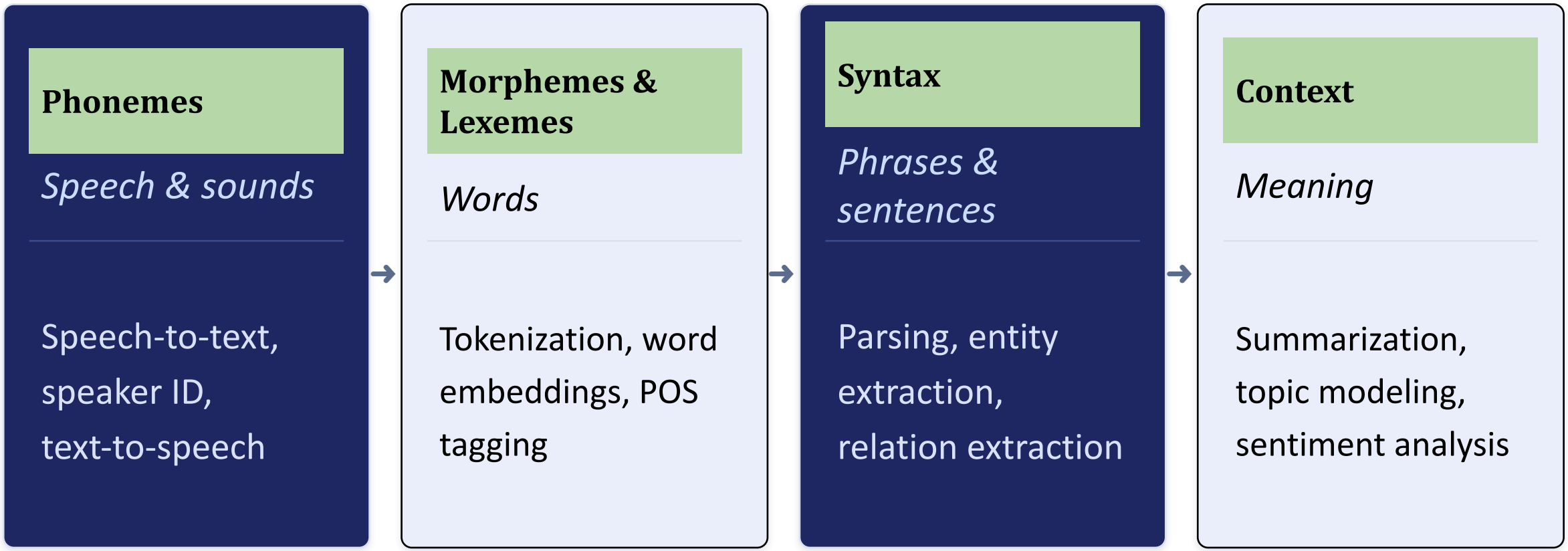
Linguistics = Understanding how language works. The scientific study of human language.

It studies things like:

- **Words** – What words mean
- **Grammar** – How words are arranged
- **Sounds** – How we pronounce words
- **Meaning** – What sentences mean
- **Context** – How the situation changes meaning

Blocks of Language

Each level gives the computer **more information about language**.



Speech & Sounds

A phoneme is a basic sound unit that can change the meaning of a word.

bat vs **pat**

bat → **/b/** pat → **/p/**

Changing /b/ to /p/ changes the whole word.

SPEECH RECOGNITION EXAMPLE

“Hello”

↓ *the computer processes the sounds*

Hello

Words

This level deals with words and their internal structure.

MORPHEME

Smallest unit of meaning.

cats = cat + s

unbreakable = un + break + able

LEXEME

The underlying word/concept and all its grammatical forms.

go, goes, went, going, gone → **GO**

run, runs, ran, running → **RUN**

Phrases & Sentences

Syntax = rules for arranging words into meaningful sentences.

"Ravi eats mangoes."

Ravi (Subject) → **eats** (Verb) → **mangoes** (Object)

Structure: Subject + Verb + Object

WORD ORDER CHANGES MEANING

"The dog chased the boy."

"The boy chased the dog."

Same words, different arrangement, different meaning.

Meaning in Situation

A word can have different meanings depending on context.

“I went to the bank to deposit money.”

bank = Financial institution

“I sat on the bank of the river.”

bank = Side of a river

All Five Blocks, One Sentence

SENTENCE

“Riya went to the bank to deposit her salary.”

BLOCK	WHAT THE COMPUTER DOES	EXAMPLE
1. Phonemes	Speech recognition converts sounds into text.	*/Riya/ /went/ /to/ /the/ /bank/…*
2. Morphemes & Lexemes	Recognizes words, forms, and lexemes.	*depositing = deposit + ing; go → went*
3. Syntax	Identifies grammatical relationships.	*Riya → went → bank*
4. Context	Uses surrounding words to disambiguate.	*“deposit her salary” → bank = financial*
5. Meaning	Understands the full intended meaning.	*Riya went to a bank to put her salary into her account.*

Understanding a Movie

A simple analogy for the five blocks of language.

Phonemes	Morphemes & Lexemes	Syntax	Context	Meaning
It first hears the sounds.	It recognises the words.	It understands how words form sentences.	It connects it with what happened before and after.	It understands what the speaker actually means.

NLP tries to move from the sounds and words people produce to the deeper meaning they intend to communicate.

Core NLP Terminology (IMPORTANT)

TERM	SIMPLE MEANING	EXAMPLE
Phonology	Study of how sounds are organized	*The difference between the sounds in "bat" and "pat"*
Morphology	Study of how words are formed	*un + happy = unhappy*
Morpheme	Smallest meaningful unit in a word	*un + happy + ness*
Syntax	How words are arranged into sentences	*"Ravi eats mangoes."*
Semantics	Meaning of words / sentences	*"The dog chased the cat" means the dog is doing the chasing*
Pragmatics	Meaning based on situation / context	*"It's cold here" may really mean "Please close the window."*
Discourse	How one sentence relates to previous sentences	*"Ravi bought a car. He loves it." → "He" refers to Ravi*
World Knowledge	General knowledge about the real world	*Knowing that people usually eat food, not cars*

The 5 Phases of NLP

The 5 Phases of NLP

1

Lexical & Morphological

Break text into words/tokens; analyze word structure.

“running” → run + ing

2

Syntactic Analysis

Check grammar & build a parse tree of sentence structure.

Subject – Verb – Object

3

Semantic Analysis

Extract the literal meaning of words & sentences.

“bank” = river edge vs. finance

4

Discourse Integration

Interpret meaning in context of surrounding sentences.

Pronoun resolution (“he”, “it”)

5

Pragmatic Analysis

Understand intended meaning based on real-world context.

“Can you pass the salt?” = request

PHASE 1· LEXICAL & MORPHOLOGICAL ANALYSIS

“What are the words, and how are they formed?”

SENTENCE

“Riya went to the bank because she needed to deposit her salary.”

TOKENS

Riya | went | to | the | bank | because | she | needed | to | deposit | her | salary

WORD STRUCTURE

depositing = **deposit** + **ing**

unhappiness = **un** + **happy** + **ness**

Lexical = What are the words? · Morphological = How are the words formed?

PHASE 2 · SYNTACTIC ANALYSIS

“How are the words grammatically connected?”

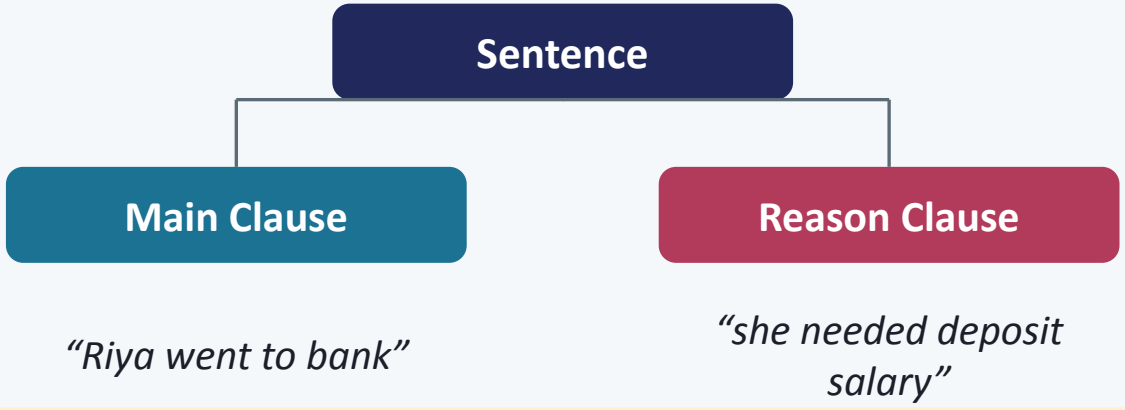
SENTENCE

“Riya went to the bank because she needed to deposit her salary.”

GRAMMATICAL ROLES

Riya	went	bank	she	needed	deposit	salary
Subject	Verb	Object / Place	Subject	Verb	Verb	Object

SENTENCE STRUCTURE



Syntactic analysis = understanding grammatical structure. The computer is essentially asking: Who did what?

PHASE 3 · SEMANTIC ANALYSIS

“What does the sentence actually mean?”

SENTENCE

“Riya went to the bank because she needed to deposit her salary.”

THE WORD “BANK” IS AMBIGUOUS — IT COULD MEAN:

Financial institution	Side of a river

Clue in context: “…needed to deposit her salary.”

bank = Financial institution — “Riya went to a financial institution to deposit money.”

Same idea applies to “The bat is on the table” — animal or sports equipment ? Semantic analysis = understanding literal meaning.

“How does this sentence connect with other sentences?”

“Riya went to the bank. She needed to deposit her salary.”

“She” = **Riya**

“Rahul bought a new laptop. It was very expensive.”

“It” = **laptop**

This is called pronoun resolution — the computer uses earlier sentences to interpret the current one.

“What does the speaker REALLY intend?”

“Can you pass the salt?”

LITERAL MEANING

A yes/no question about ability.

REAL INTENTION

Please pass me the salt.

“It’s very noisy in here.”

LITERAL MEANING

A statement about the noise level.

REAL INTENTION

Please be quiet.

Pragmatic analysis = understanding what the speaker actually intends in the real-world situation.

EXAMPLE

SENTENCE

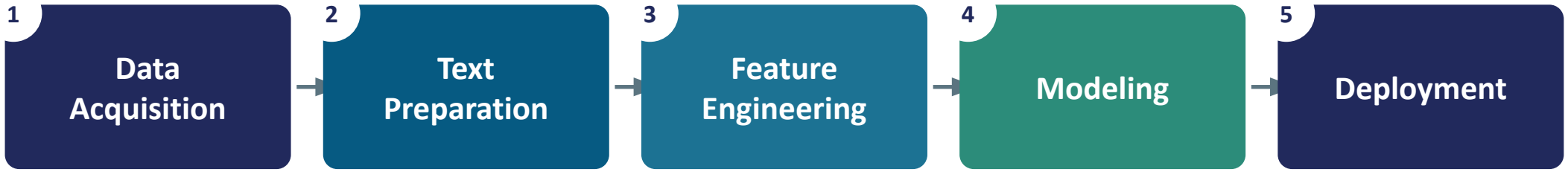
“Rahul opened the window because it was very hot.”

1	**Lexical**	Rahul | opened | the | window | because | it | was | very | hot
2	**Syntactic**	Rahul → opened → window
3	**Semantic**	“opened the window” = made the window open
4	**Discourse**	“It” = the surrounding situation / room, from context
5	**Pragmatic**	Real intention: he wanted fresh air / cooling because the room was hot — not just to touch the window

End-to-End NLP Pipeline

What Is an NLP Pipeline?

An NLP pipeline is the set of ordered steps followed to design, build, and ship a complete Natural Language Processing software solution from raw, messy and unstructured text.



The Five Pipeline Stages

1	**Data Acquisition**	Collect or source the raw text data needed for the project.
2	**Text Preparation**	Cleaning → Basic Preprocessing → Advanced Preprocessing.
3	**Feature Engineering**	Convert cleaned text into numeric features a model can use.
4	**Modeling**	Build the model (Heuristic / ML / DL / Cloud API) & evaluate it.
5	**Deployment**	Deploy → Monitor → Update the model in production.

Data Acquisition

The first step of any NLP project: obtaining the **textual data needed to train and build the model.** Imagine your manager asks you to build a sentiment-analysis system — where does the data come from?

1 Data Already Available

Your organization already has labeled/unlabeled text. Just preprocess it and start.

2 Another Team/Company Has It

Data exists elsewhere. Coordinate & request access to the relevant dataset.

3 No Data Exists Anywhere

The hardest case — you must build the dataset yourself from scratch.

If no data exists — how do you build it?

Web Scraping

Extract text from websites/forums

Public Datasets

Kaggle, HuggingFace, government data

APIs

Twitter/X, Reddit, news APIs, etc.

PDFs / Images / Audio

Extract via OCR & speech-to-text

Text Preparation – Text Cleaning

Strips out noise that carries no linguistic value and would confuse downstream steps.

Remove HTML Tags

Great product!

↓

Great product!

Remove Emojis / Emoticons

Loved it

↓

Loved it

Spelling Correction

Ths is amazng

↓

This is amazing

Why it matters: Web-scraped or social-media text is full of markup, emoji, typos & inconsistent casing. Cleaning it early prevents these artifacts from being mistaken for meaningful tokens later in the pipeline.

Text Preprocessing Pipeline

Text preprocessing means **cleaning and preparing raw text** by removing unwanted or unnecessary parts so that a computer can easily analyze it.

Example:

```
"I LOVE this movie!!!"
"love movie"
```

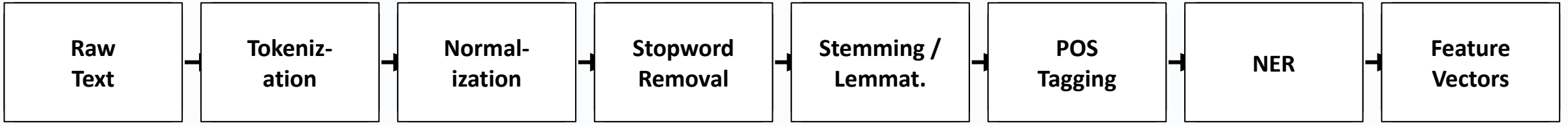
It Means

Raw Text → Clean Text → Ready for Analysis

Data Preprocessing Pipeline

Stage 2 of the NLP Pipeline

Before any NLP model can learn from text, raw language must be cleaned and structured through a series of preprocessing steps.



STEP	WHAT IT DOES
Tokenization	Split text into words/sentences.
Normalization	Lowercase, remove punctuation/noise
Stemming/Lemmatisation	Reduce words to their root form
POS Tagging	Label the grammatical role of each word.
NER	Identify names, places, dates, organizations

Tokenization

What & Why: Splitting text into smaller units — words, subwords, or sentences — called tokens. It is usually the first step of any NLP pipeline.

EXAMPLE

"NLP is transforming how machines understand language!"

Word tokens:

NLP | is | transforming | how | machines | understand | language | !

Sentence tokens: 1 sentence total

Types: Word tokenization (split on spaces/punctuation) · Sentence tokenization (split on sentence boundaries) ·

REAL-LIFE EXAMPLE

Search engines tokenize your query before searching their index — "best pizza near me" becomes 4 separate tokens the engine can match against web pages.

Text Normalization

What & Why: Converts text into a single, consistent, standard form — reducing noise and variation so the model treats equivalent text the same way.

Lowercasing

"NLP" → "nlp"

Removing Punctuation

"Wow!!!" → "Wow"

Expanding Contractions

"don't" → "do not"

Removing Extra Whitespace

"hi there" → "hi there"

Removing Stopwords

"the cat is on the mat" → "cat mat"

Spelling Correction

"recieve" → "receive"

BEFORE

"Don't U think it's AMAZING?? :) "

→

AFTER

"do not you think it is amazing"

Stemming (lexicon normalization)

What & Why: Reduces a word to its root/stem by chopping off prefixes or suffixes.

```
from nltk.stem.wordnet import WordNetLemmatizer
lem = WordNetLemmatizer()

from nltk.stem.porter import PorterStemmer
stem = PorterStemmer()
```

Original Word	Stemmed Form	Rule Applied
running	**run**	*remove “ning”*
studies	**studi**	*remove “es”*
happiness	**happi**	*remove “ness”*
organization	**organ**	*remove “ization”*
easily	**easili**	*remove “ly”*

STEMMING vs. LEMMATIZATION

Stemming is faster but cruder (“studies” → “studi”).

Lemmatization uses vocabulary & grammar rules to return a real word (“studies” → “study”), but is slower.

Key takeaway: choose stemming for speed, lemmatization for accuracy.

Parts-of-Speech (POS) Tagging

What & Why: Assigns a grammatical category — noun, verb, adjective, etc. — to each word based on its meaning and context in the sentence.

The	quick	fox	jumps	over	the	lazy	dog
DET	ADJ	NOUN	VERB	PREP	DET	ADJ	NOUN

DET = Determiner · ADJ = Adjective · NOUN = Noun · VERB = Verb · PREP = Preposition

a word that comes before a **noun** and tells us **which one, whose one, or how many**

WHY IT MATTERS

POS tags resolve ambiguity (“book” as NOUN vs. VERB), and feed into parsing, NER, and grammar-checking systems.

COMMON TAG SET

Noun, Pronoun, Verb, Adjective, Adverb, Preposition, Conjunction, Determiner, Interjection

Named Entity Recognition

How machines learn to find people, places, organizations, and dates hidden inside plain text?

What is NER?

Named Entity Recognition is an NLP technique that finds important names or **information in a sentence and classifies them into categories** — turning unstructured text into structured data.

WHERE IT'S USED

- **Search**

  Surface people, places & orgs instantly

- **Chatbots**

  Resolve names, locations, and dates

- **Summarization**

  Anchor summaries around key entities

- **Knowledge Graphs**

  Link entities and their relationships

EXAMPLE

Riya bought a laptop from Amazon on 5th August 2026.

Riya	**Person**
Amazon	**Organization**
5th August 2026	**Date**

Unstructured sentence → Structured entity data

Common Entity Types

Person	Organization	Location	Date
Albert Einstein, Riya	Amazon, Google	Paris, Mumbai	5th August 2026

Time	Money	Percentage
10:30 AM	₹50,000, $100	50%, 25%

Same Word, Different Entity

NER doesn't just look at a word — it reads the sentence around it (context).

AMAZON

Amazon is expanding rapidly.

Context clue

“expanding rapidly”

ORGANIZATION

AMAZON

The Amazon is the largest rainforest.

Context clue

“largest rainforest”

LOCATION

Key idea: Meaning + Context → Correct Entity

How Does NER Work?

1

Text Input

System receives a sentence

→

2

Segmentation

Text split into sentences

→

3

Tokenization

Sentence split into tokens

4

POS Tagging

Grammatical role identified

→

5

Entity Detection

Candidate entities located

→

6

Classification

Each entity gets a category

Result: Riya → PERSON Amazon → ORGANIZATION

Four Approaches to NER

1. Lexicon-Based

Uses a dictionary of known entities (e.g. Mumbai, Delhi, Pune). Fast, but misses unknown names.

2. Rule-Based

Applies handcrafted patterns — e.g. a word after “Dr.” is likely a person. Needs manual upkeep.

3. Machine Learning

Learns from labeled data using classifiers or Conditional Random Fields (CRF) over word sequences.

4. Deep Learning

BiLSTM, CNN, and Transformer models learn context automatically from word embeddings.

Putting It All Together

Shreyansh visited Symbiosis University in Pune on 12 August 2026 and paid ₹50,000.

PERSON	ORGANIZATION	LOCATION	DATE	MONEY
Shreyansh	Symbiosis University	Pune	12 August 2026	₹50,000

Find important words → understand their type → convert text into structured information.

A Simple Implementation — spaCy

```
import spacy

nlp = spacy.load("en_core_web_sm")

text = "Siya joined Google in Mumbai"
        " on 5th August 2026."

doc = nlp(text)

for entity in doc.ents:
    print(entity.text, "→", entity.label_)
```

OUTPUT

Siya	→	**PERSON**
Google	→	**ORG**
Mumbai	→	**GPE**
5th August 2026	→	**DATE**

```
pip install spacy && python -m spacy
download en_core_web_sm
```

DATA PREPROCESSING

Stage 2 of the NLP Pipeline

Feature Engineering

Machine learning models can't read words — they need numbers. Feature engineering converts cleaned text into meaningful numeric representations.

Bag of Words

Matrix of word counts per document; ignores word order.

TF-IDF

Weighs words by frequency in a doc vs. rarity across the corpus.

Word Embeddings

Dense vectors (Word2Vec, GloVe) capturing semantic similarity.

N-grams

Captures short word sequences to preserve local order.

POS Tags as Features

Grammatical roles feed into NER & parsing models.

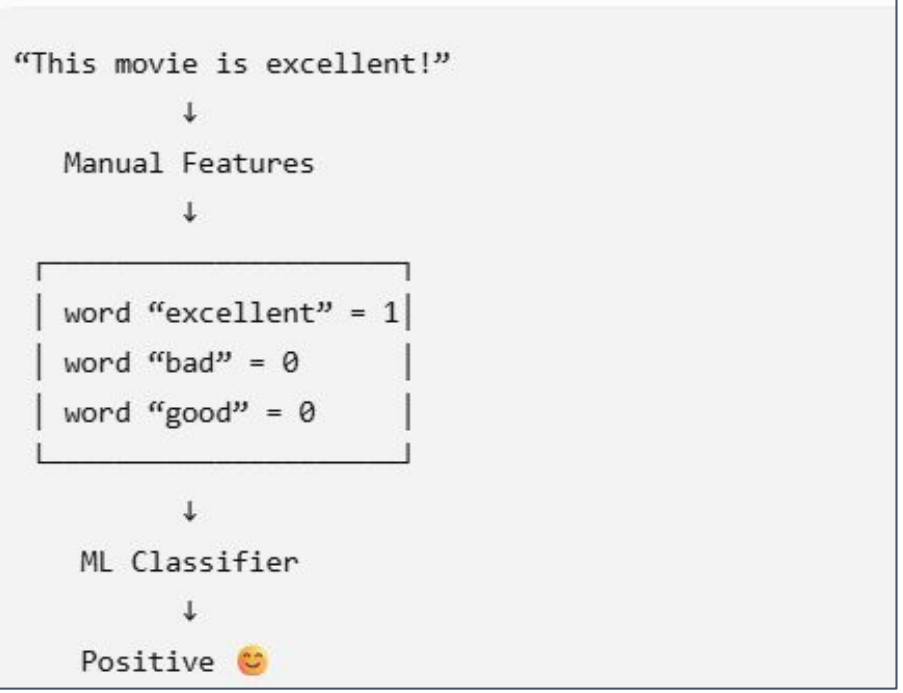
Dependency Parse Features

Grammatical relationships help extraction & QA tasks.

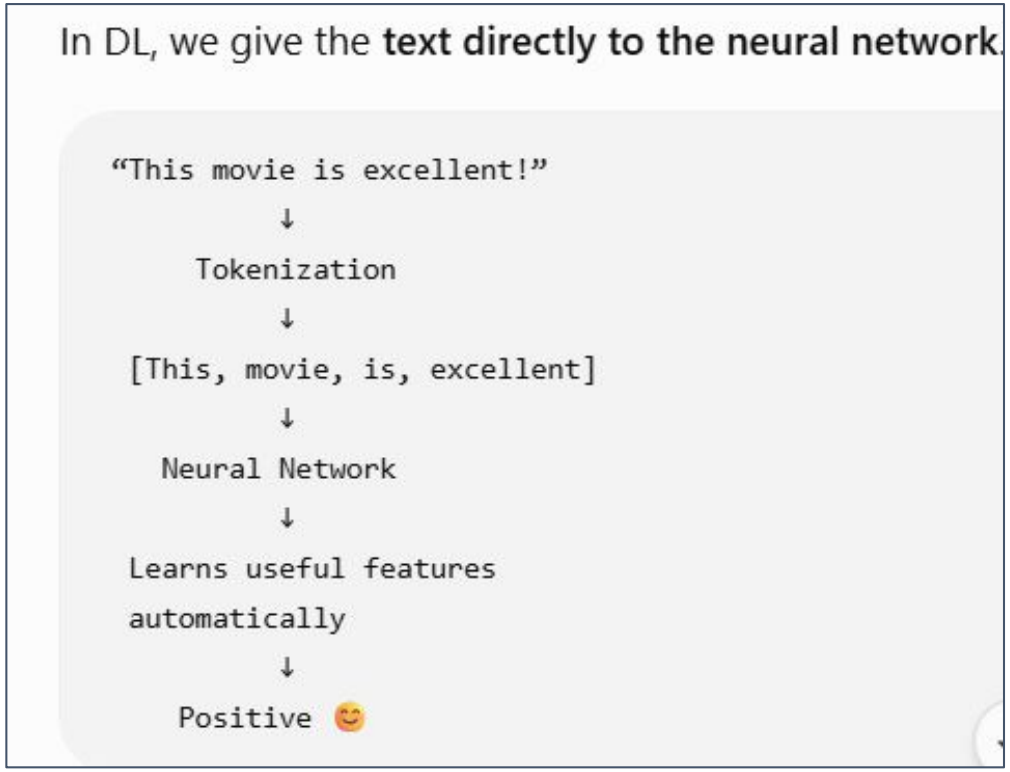
ML vs. DL: In classical ML, features are engineered manually (as above). In deep learning, the network learns its own features automatically from raw/tokenized text

Feature Engineering

In ML, **we manually create features** from the text.



In DL, we give the **text directly to the neural network**.



The programmer decides **which features** are important.

For example:

- Number of positive words
- Number of negative words
- Presence of words like *excellent, good, bad, terrible*

The neural network **learns which words, patterns, and relationships are important** during training.

FEATURE EXTRACTION IN NLP **DATA PREPROCESSING** **Stage 2 of the NLP Pipeline**

Turning Text Into Numbers

Feature extraction converts text into useful numerical information that a machine-learning model can understand.

😊 **"I love this movie."**

😢 **"I hate this movie."**

WORD FREQUENCY

WORD	SENTENCE 1	SENTENCE 2
I	1	1
love	1	0
hate	0	1
movie	1	1

SENTENCE 1 →

```
[ 1, 1, 0, 1 ]
```

SENTENCE 2 →

```
[ 1, 0, 1, 1 ]
```

The model learns: "love" → positive sentiment, "hate" → negative sentiment.

Feature Extraction = Text → Useful Numerical Features → ML Model

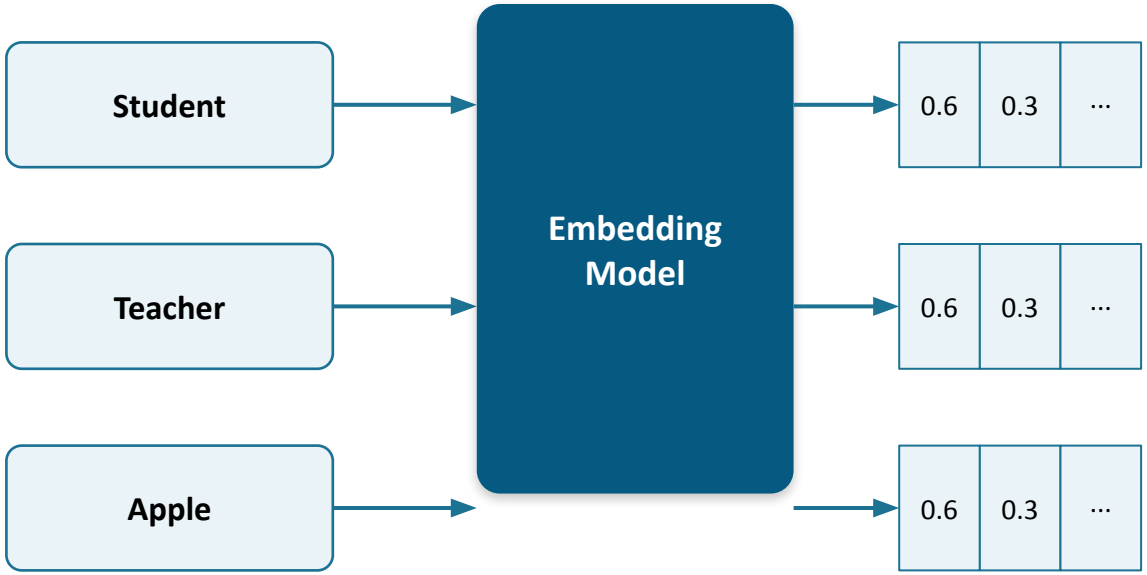
Word Embeddings

Turning Words into Numbers

What Is a Word Embedding?

Computers can't read words — they only work with numbers.

So in NLP, every word is converted into a list of numbers called a **vector**. That numerical representation is the word embedding.



Objects → Model → Vectors

Word Embedding =

Representing a word as a list of numbers (a vector) that captures its meaning.

What Is a Vector?

A vector is simply a list of numbers. Each number is one dimension of the word's representation.

4-DIMENSIONAL EXAMPLE

0.62	**0.81**	**-0.12**	**0.45**

"student" (each value represnets features its 4 dimension)

REAL MODELS USE FAR MORE DIMENSIONS

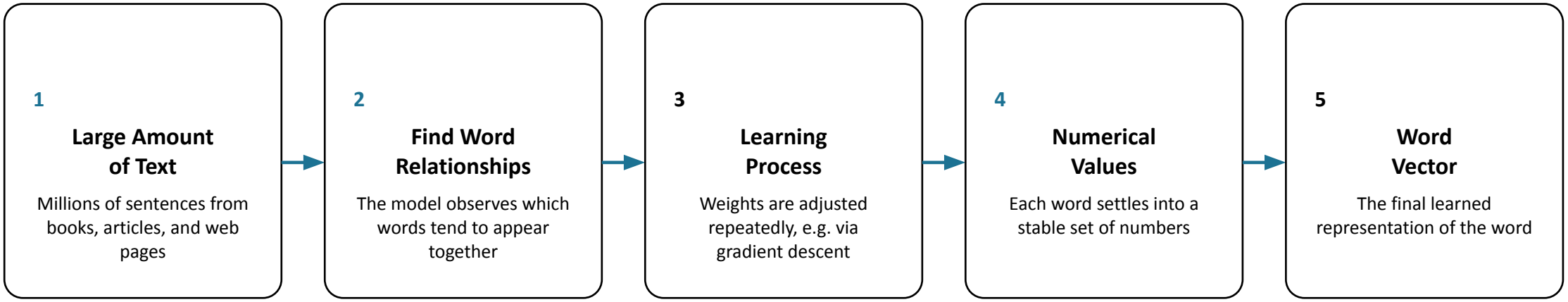
50–300	**768**	**1000s**
Word2Vec / GloVe	BERT	Large LLM embeddings

Why only 4 dimensions in the example?

Four is simply easy to read on a slide. The number of dimensions is a design choice fixed by the model's architecture — it carries no special meaning on its own.

How Are These Values Generated?

The numbers are never assigned by hand — they're learned automatically by a model that studies huge amounts of text.



EXAMPLE OUTPUT

student → `[ 0.62, 0.81, -0.12, ... ]`

teacher → `[ 0.60, 0.79, -0.10, ... ]`

car → `[ 0.10, -0.40, 0.75, ... ]`

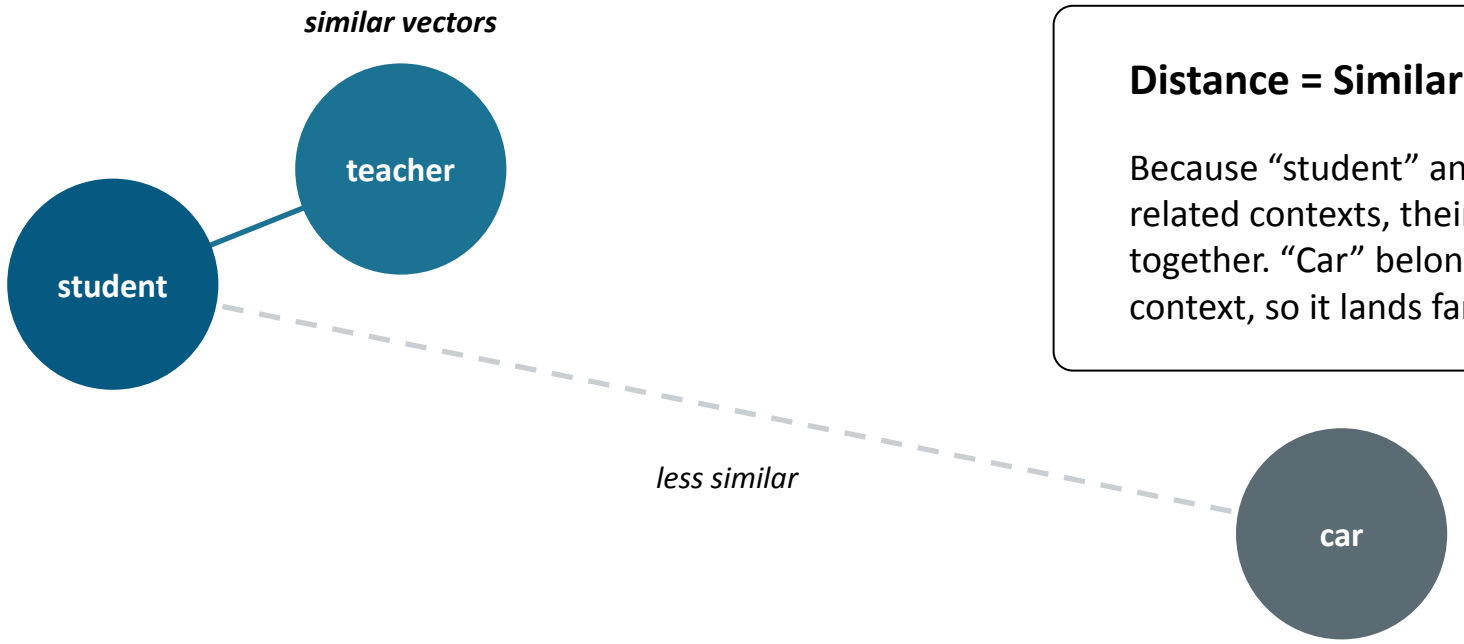
A Closer Look: Training One Word

Tracing how the vector for "student" evolves from nothing to something meaningful.

Model Reads Many Sentences	→	**Starts With Random Numbers**	→	**Training Adjusts the Numbers**
"The student studies in college." *"The teacher helps the student."*		`student → [0.15, -0.32, 0.71, 0.08]`		Wrong predictions ⟶ weights updated (e.g. gradient descent), repeated millions of times.
"student" keeps appearing near college, class, teacher.		These numbers hold no meaning yet.		**`student → [0.62, 0.81, -0.12, 0.45]`**

Why Are Similar Words Close?

Words used in similar contexts end up with similar vectors — so distance between vectors becomes a measure of meaning.



Distance = Similarity

Because “student” and “teacher” appear in related contexts, their vectors sit close together. “Car” belongs to a different context, so it lands far away.

This is the core idea behind semantic search, recommendation engines, and analogy tasks like “king − man + woman ≈ queen.”

Main Types of Word Embeddings

Frequency-Based

Counts words

EXAMPLES

BOW · TF-IDF

Represents words using raw or weighted counts of how often they occur.

Prediction-Based

Learns from surrounding words

EXAMPLES

Word2Vec · GloVe · FastText

Trains a model to predict context, learning dense vectors along the way.

Contextualized

Meaning shifts with sentence context

EXAMPLES

ELMo · BERT · GPT

The same word gets a different vector depending on how it's used.

TEXT PREPROCESSING SUMMARY

Stage 2 of the NLP Pipeline

From Raw Text to Model-Ready Features

ORIGINAL TEXT

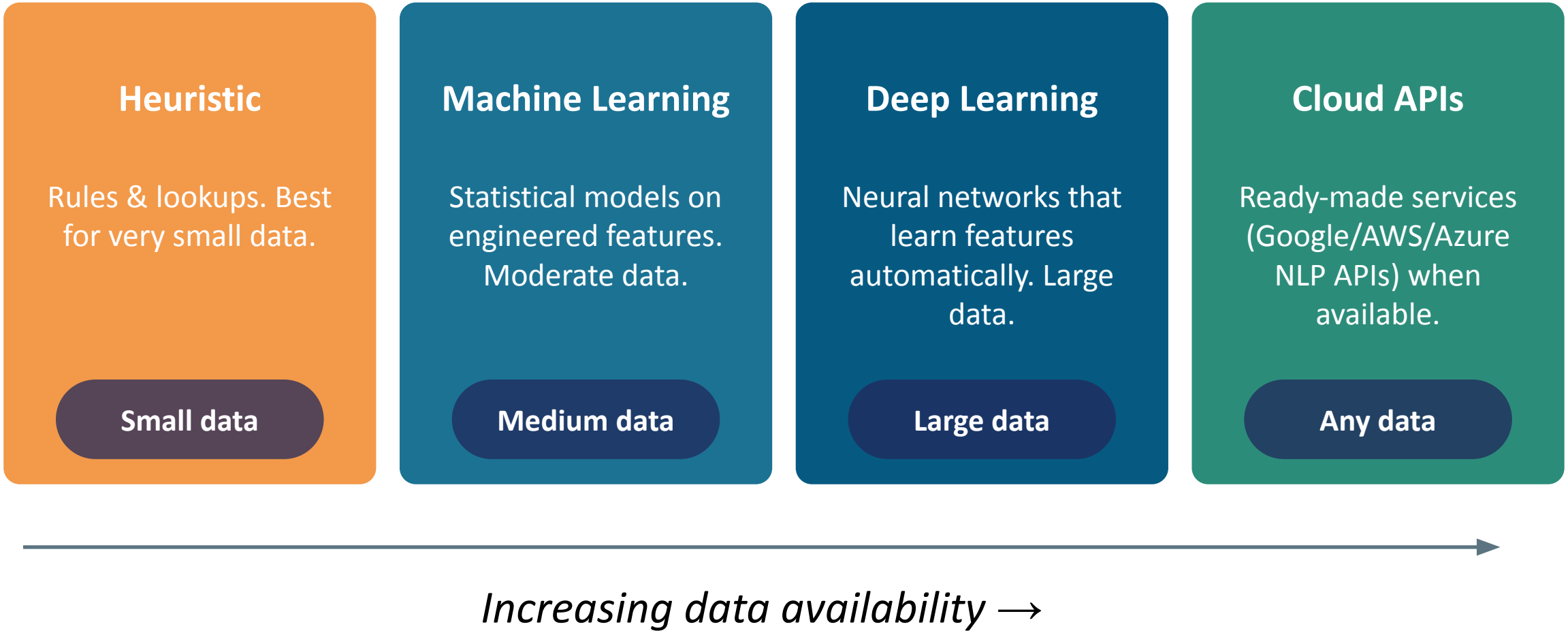
“The boys are running!!! They love playing football. 😊”

PREPROCESSING TASK	PURPOSE	EXAMPLE (APPLIED TO THE SENTENCE)
1. Noise Reduction	Removes punctuation, symbols, and unwanted text.	`“The boys are running They love playing football”`
2. Normalization	Converts different word forms into a standard form.	`“boy run love play football”`
3. Tokenization	Splits text into words or smaller units.	`["boy", "run", "love", "play", "football"]`
4. Stopword Removal	Removes common words like the, is, and.	`Remove words like the, are, they`
5. Feature Extraction	Converts text into useful numerical features.	`Convert words into numbers / features`
6. Dimensionality Reduction	Reduces the number of features to simplify analysis.	`Keep only the most useful features`

Modeling — Four Approaches

Stage 3 of the NLP Pipeline

Modeling means building the model and evaluating it. Which of the four approaches usually comes down to how much data is there.



Model Evaluation

Stage 4 of the NLP Pipeline

Checks how well the model works on unseen data.

Intrinsic Evaluation (How well the model performs)

Measures the model's performance on its specific NLP task.

Examples: Precision, Recall, F1, BLEU, Perplexity

Extrinsic Evaluation (How useful the model is in real life)

Measures how well the model works in real-world use.

Example: User satisfaction, task success, usage rate

Precision: Measures how many predicted positive results are actually correct.

Recall: Measures how many actual positive results the model correctly finds.

F1 Score: Combines Precision and Recall into one balanced score.

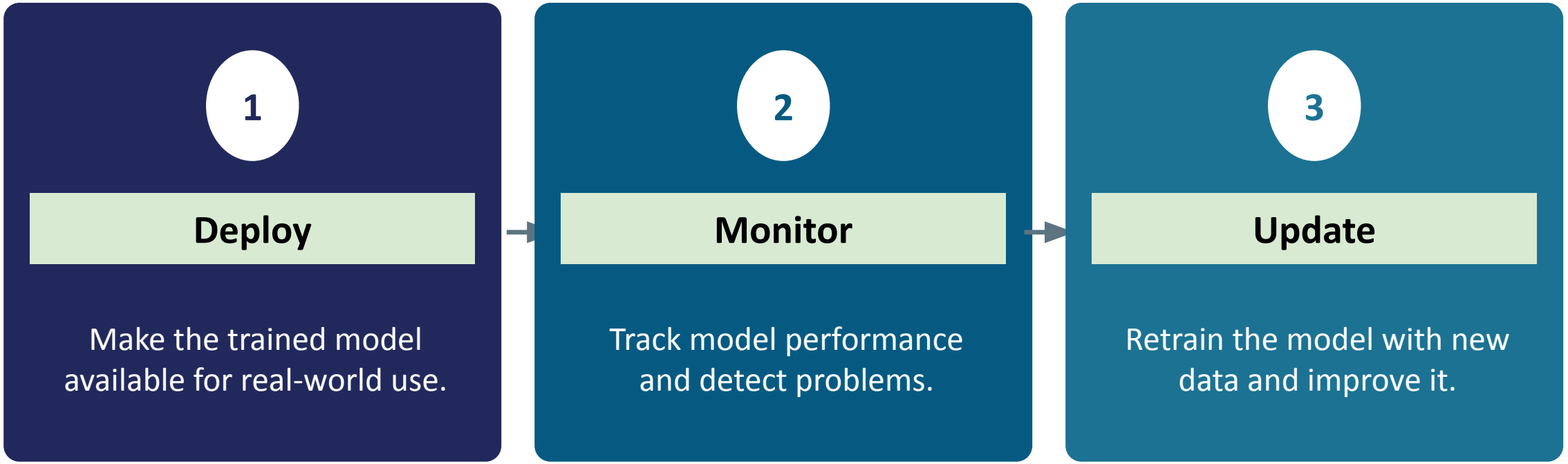
BLEU: Measures how closely a machine-generated translation matches a human translation.

Perplexity: Measures how well a language model predicts the next word; lower is better.

Deployment

Stage 5 of the NLP Pipeline

Making a trained model available for real use — integrated into an application so it can serve live predictions.



This is a loop, not a one-time event

Information Retrieval vs Information Extraction in NLP

Information Retrieval vs Information Extraction

Two NLP tasks that sound alike but do very different jobs

Information Retrieval (IR)

Finding the most relevant documents from a large collection.

Query: *"Best Python courses for beginners"*

A search engine scans millions of pages and retrieves the most relevant ones.

Query → Search → Relevant documents

Information Extraction (IE)

Pulling specific facts out of text and turning them into structured data.

Text: *"Ravi joined Google in Mumbai in 2024."*

Person: Ravi **Org:** Google **Location:** Mumbai **Year:** 2024

Text → Identify key info → Structured data

Main Difference

	Information Retrieval	Information Extraction
Goal	Find relevant documents	Extract useful facts
Input	A query	Text / document
Output	Documents / pages	Structured information

Note:

IR = "Find it"

IE = "Take it out"

Google finds the page (IR); IE pulls the name, place, date out of it.

Example

“Apple CEO Tim Cook visited Mumbai on Monday.”

IR — Retrieval

Find documents about Tim Cook's visit to Mumbai.

A search engine returns news articles, reports, and pages that discuss this visit — it does not pick apart the sentence itself.

IE — Extraction

Person	**Tim Cook**
Organization	**Apple**
Location	**Mumbai**
Date	**Monday**

Information Retrieval — Techniques

Boolean Search

Finds documents using AND, OR, NOT.

Cosine Similarity

Measures similarity between query and document.

TF-IDF

Finds important words in documents.

Word Embeddings

Finds documents with similar meaning.

BM25

Ranks documents based on relevance.

BERT / Transformers

Understands query and document context.

Vector Space Model

Represents documents as vectors.

Semantic Search

Finds information based on meaning, not just keywords.

Information Extraction — Techniques

Named Entity Recognition (NER)

Extracts names, places, organizations, dates, etc.

Pattern Matching / Regex

Extracts information using predefined patterns.

Part-of-Speech (POS) Tagging

Identifies nouns, verbs, adjectives, etc.

Rule-Based Extraction

Uses manually defined language rules.

Relation Extraction

Finds relationships between entities.

Machine Learning

Learns extraction patterns from examples.

Keyword Extraction

Identifies important words or phrases.

Deep Learning / Transformers

Extracts complex information using models such as BERT.

Thank You

Next-Word Prediction in WhatsApp & Email

When you type:

“I am going to…”

The system uses a **Neural language model** to understand the context and predict the most likely next word.

Process:

Typed words → Context analysis → Probability calculation → Next-word prediction

Example:

I am going to → school / office / market

Modern systems use **Deep Learning** and **Transformer-based language models** for better predictions

Transformer Model & Self-Attention

Transformer is a deep learning architecture widely used in NLP for tasks like next-word prediction, translation, summarization, and chatbots.

Self-Attention is the key mechanism inside a Transformer. It helps the model understand which words in a sentence are important to each other.

Example:

> **"The boy put the book on the table because it was heavy."**

Self-attention helps determine what **"it"** refers to by looking at the surrounding words and their relationships.

flow:

Input sentence → Tokenization → Self-Attention → Context understanding → Prediction

Self-Attention = Understand relationships between words

Transformer = Architecture that uses self-attention to process language efficiently

NLP Sentence Analysis – Questions

Tokenization | POS Tagging | NER | Sentiment Analysis | Discourse & Dialog Analysis

Sentence 1

"Riya bought a new laptop from Amazon yesterday."

1. Identify all the tokens.
2. Identify the nouns and verbs.
3. Identify the named entities and their types.
4. Which word represents a location, if any?
5. Identify the person and organization/product entities.

Sentence 1

"Riya bought a new laptop from Amazon yesterday."

1. **Identify all the tokens.**
   Ans: Riya, bought, a, new, laptop, from, Amazon, yesterday, .
2. **Identify the nouns and verbs.**
   Ans: Nouns: Riya, laptop, Amazon, yesterday. Verb: bought.
3. **Identify the named entities and their types.**
   Ans: Riya → PERSON; Amazon → ORGANIZATION; yesterday → DATE/TIME.
4. **Which word represents a location, if any?**
   Ans: No location entity is present in this sentence (Amazon here refers to the company, not the river/rainforest).
5. **Identify the person and organization/product entities.**
   Ans: Person: Riya. Organization: Amazon. Product (generic, not a named entity): laptop.

“I ordered three books online, but they arrived damaged.”

33. Identify the tokens.
34. Identify the pronoun.
35. What does “they” refer to?
36. Identify the negative opinion.
37. Identify the noun and verb.
38. Which NLP technique helps resolve “they”?

“I ordered three books online, but they arrived damaged.”

33. Identify the tokens.

Ans: I, ordered, three, books, online, but, they, arrived, damaged, .

34. Identify the pronoun.

Ans: I and they.

35. What does “they” refer to?

Ans: The three books.

36. Identify the negative opinion.

Ans: “they arrived damaged” → negative opinion about the product condition.

37. Identify the noun and verb.

Ans: Noun: books. Verbs: ordered, arrived.

38. Which NLP technique helps resolve “they”?

Ans: Coreference resolution (anaphora resolution).